SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1712**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I R. Sandhya Rani (192372378) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

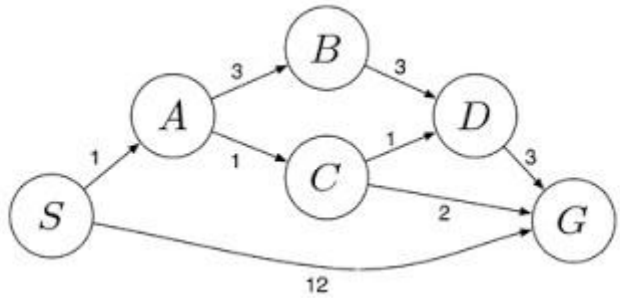
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. **Water Jug Problem (4-Gallon and 3-Gallon Jugs)**

Problem Statement

You are given:

- One **4-gallon jug**
- One **3-gallon jug**
- A water pump with an unlimited supply of water

Neither jug has any measurement markings. The allowed operations are:

1. Fill a jug completely from the pump.
2. Empty a jug onto the ground.
3. Pour water from one jug into the other until either:
   - the source jug becomes empty, or
   - the destination jug becomes full.

Goal: Obtain **exactly 2 gallons of water in the 4-gallon jug.**

Final Answer

By following the sequence:

(0,0) → (0,3) → (3,0) → (3,3) → (4,2) → (0,2) → (2,0)

we successfully obtain **exactly 2 gallons of water in the 4-gallon jug**.

Goal Achieved: 4-gallon jug = 2 gallons, 3-gallon jug = 0 gallons.

2. **Mars Rover: AI Agent Analysis (Short Answer)**

i. What are the percepts for this agent?

Percepts are the inputs received through the rover's sensors:

- Camera images
- Terrain and obstacle information
- Rock and soil composition
- Temperature and weather data
- Position and orientation
- Battery level and system health

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ii. Characterize the operating environment

- **Partially Observable** – Cannot see the entire environment.
- **Stochastic** – Outcomes are uncertain due to terrain and weather.
- **Sequential** – Current actions affect future tasks.
- **Dynamic** – Environment changes over time.
- **Continuous** – Movement and sensor readings are continuous.
- **Single-Agent** – Operates independently.

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iii. What actions can the agent take?

- Move and navigate

- Avoid obstacles
- Capture images
- Collect and analyze rock/soil samples
- Send data to Earth
- Monitor power and system status

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iv. How can the performance be evaluated?

The rover is evaluated based on:

- Successful mission completion
- Accurate sample analysis
- Safe navigation
- Efficient energy usage
- Reliable communication with Earth

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v. Which agent architecture is most suitable? Why?

Utility-Based Learning Agent

Reason: It can learn from experience, adapt to changing conditions, make intelligent decisions, and maximize mission success while minimizing risks and power consumption.

3. 8-Queens Problem – Search Space Formulation

Problem Statement

The **8-Queens problem** requires placing **8 queens** on an **8 × 8 chessboard** so that **no two queens attack each other**. A queen can attack another queen if they are in the **same row, same column, or same diagonal**.

Problem Components

1. Initial State

- An **empty 8 × 8 chessboard** with no queens placed.

2. State Space

- All possible arrangements of queens on the chessboard.
- A state represents a board configuration with **0 to 8 queens** placed.

3. Operators (Actions)

- Place one queen in a safe position on the next row.
- Move a queen if a conflict occurs (during backtracking).

4. Goal State

- Place all **8 queens** on the board such that:
  - No two queens are in the same row.
  - No two queens are in the same column.

- No two queens are on the same diagonal.

5. Path Cost

- Each queen placement is considered one step.
- The objective is to reach a valid solution with the fewest backtracking steps.

Search Strategy

The **Backtracking Search (Depth-First Search)** is the most commonly used strategy.

Performance of the Search

- **Search Method:** Backtracking (Depth-First Search)
- **Completeness:** Yes, it finds a solution if one exists.
- **Time Complexity:** Approximately **O(8!)**
- **Space Complexity:** **O(8)** (stores one queen position per row).

Conclusion

The **8-Queens problem** is a classic AI search problem. It is formulated using an **initial state**, **state space**, **operators**, **goal state**, and **path cost**. The most suitable search strategy is **Backtracking (Depth-First Search.**

4. OLA Cab Booking – Problem Solving Agent
Type of Problem-Solving Agent
The **Goal-Based Agent** is the most suitable for this problem.
Reason:
The customer's goal is to **reach the destination**. The agent evaluates different cab options (Mini, Micro, Sedan, Prime, Shared, etc.) based on factors like **cost, comfort, and availability**, then selects the most suitable cab to achieve the goal.

Pseudocode

```
BEGIN

Input source location
Input destination location

Display available cab types
(Mini, Micro, Sedan, Shared, Prime)

Customer selects a cab type

IF selected cab is available THEN
    Book the cab
    Display driver details
```

```
    Display fare and estimated arrival time
    Start the trip
ELSE
    Display "Selected cab is not available"
    Suggest another available cab
ENDIF

END
```

Working Steps

1. Customer enters the pickup and destination locations.
2. The app displays available cab types.
3. The customer selects the preferred cab.
4. The system checks cab availability.
5. If available, the cab is booked and trip details are shown.
6. Otherwise, the app suggests another available cab.

Conclusion

A **Goal-Based Agent** is appropriate because it selects actions that help the customer achieve the goal of reaching the destination efficiently while considering the chosen cab type and its availability.

5. Uniform Cost Search (UCS)

Step 1: Edge Costs

- S → A = 1
- S → G = 12
- A → B = 3
- A → C = 1
- B → D = 3
- C → D = 1
- C → G = 2
- D → G = 3

Step 2: UCS Execution

Step	Node Expanded	Cost
1	S	0
2	A	1
3	C	2
4	D	3
5	G	**4 (Goal)**

Least-Cost Path

S → A → C → G

Total Cost = 1 + 1 + 2 = 4

Pseudocode

```
BEGIN
Insert S into priority queue

WHILE queue is not empty
    Remove node with lowest cost
    IF node = G
        Return path and cost
    Expand node
    Add child nodes with cumulative cost
END WHILE

END
```

Final Answer

- **Algorithm:** Uniform Cost Search (UCS)
- **Optimal Path: S → A → C → G**
- **Minimum Cost: 4**